

Sub-theme:

Project-based learning activities emphasizing IKS principles

Integrating Indian Knowledge Systems into Project-Based Learning for Enhancing Life Satisfaction in Indian Construction Companies: A Pilot Study

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Abstract

This pilot study explores the integration of Indian Knowledge Systems (IKS) into Project-Based Learning framework for enhancing life satisfaction among professionals in Indian construction companies. The research examines multiple dimensions of well-being with 19 respondents for the pilot study, including physical and mental well-being, relationship and communication skills, purpose and motivation at work, skills for personal and professional growth, financial and time management, spiritual growth, and temperament control. The study also assesses behavioral practices because of a structured intervention training imparted to the respondents such as adherence to diet and workout plans, practicing creative pauses, experiences of sharing, weight management actions, self-reflection on behavioral changes, and maintaining a daily appointment with self. The structured intervention of 8 hours for one full-day, grounded in traditional Indian wisdom and holistic living principles, was implemented to analyze its impact on respondents' lives. Data were collected through online survey questionnaire for self-reported reflections after the completion of the structured intervention and after a gap of minimum 90 days of its practice. The findings indicate that the adoption of IKS-based practices fosters increased self-awareness, self-discipline and regularity in life. The study contributes to the understanding of how indigenous knowledge can be systematically integrated into modern corporate environments to improve employee well-being. Future research will expand on these findings with a larger sample size and cross-sectional analysis to assess the long-term benefits of IKS-driven interventions.

Keywords: Indian Knowledge Systems (IKS), Life Satisfaction, Well-being, Construction Professionals, Project-Based Learning.

1. Introduction

Indian Knowledge Systems (IKS) contain a huge collection of numerous traditional wisdom alongside traditional knowledge and practices which originated across millennia throughout India. IKS emerged from ancient literary masterpieces including Vedas, Upanishads, and Arthashastra. Ethics based governance, Vastu Shastra, Ayurveda and Yoga serve as its main disciplines. IKS offers professionals multiple benefits that improve decision quality and achieve sustainable growth while promoting holistic well-being. The integration of IKS principles with contemporary professional frameworks enables industries to use indigenous sustainability practices and ethical leadership approaches for developing greater human motivation and understanding of well-being.

Project-Based Learning (PBL) stands as an educational methodology which teaches students by assigning practical hands-on projects as the core focus. Students develop problem-solving abilities and build critical thinking competencies to work cohesively on reflective tasks that mirror practical scenarios. Learners improve their capabilities in navigating complex professional spaces through PBL because the approach requires them to bring theoretical understanding to real-world situations. Well-being benefits from IKS alongside PBL helps participants develop satisfaction from their work achievements while creating deep commitment to their tasks such that the learning remains continuous to drive innovation.

Indian construction sites are characterized to create three significant conditions that negatively affect workers physical and mental state through the high-pressure environments, long working hours, and security threats. The modern advancements in technology and workplace management cannot eliminate the ongoing problems of workplace stress, job insecurity, and dangerous conditions at construction works. The fundamental necessity of addressing well-being in Indian construction companies is expected to lead to improved efficiency and life satisfaction in a sustainable development of the industry. By integrating, cultural heritage from India along with contemporary performance models businesses can build workplaces that are both productive and compassionate.

2. Literature Review

The paradigm of IKS delivers complete well-being through its fusion of physical well-being with mental, social elements and spiritual wellbeing. The system of Ayurveda delivers balance in physical and mental processes through its philosophy in bodily functions, while Yoga practice delivers mind control and reduction of stress along with mental resilience. Through Indian philosophical teachings of the Bhagavad Gita, people should practice self-discipline while maintaining ethical behavior to achieve peaceful inner states. Research has shown that corporate and industrial workplaces benefit when IKS-based wellness practices are to be implemented because they produce better employee health, job satisfaction and higher productivity rates.

Research shows that PBL techniques successfully build professional training expertise while improving memory retention. The combination of engineering, management and healthcare education proves that PBL builds problem-solving abilities while enabling teamwork as well as adaptability skills. Due to its implementation in corporate training programs, PBL demonstrates improved employee engagement together with better innovation results and stronger leadership capability. According to Kolb and Kolb (2006), the experiential learning cycle of PBL demonstrates that it helps develop employees' understanding and practical abilities to increase their job performance.

Human motivation and performance particularly depend on life satisfaction alongside well-being. Life satisfaction increases when people meet their psychological and social requirements according to Maslow's Hierarchy of Needs (Maslow, 1943) and Self-Determination Theory (Deci & Ryan, 1985). According to Diener's (1984) subjective well-being model, people find happiness through the combination of positive feelings with personal self-evaluations and their life's meaning. The combination of IKS and PBL methodologies with these theoretical frameworks improves both professional job satisfaction and work-life balance and resilience in work environments.

The subject matter of "Organizational Psychology", and "Human Resource Management" together provide a variety of behavioral change interventions that have structured approaches. Mindfulness training combined with cognitive-behavioral techniques and leadership development initiatives establish programs that drive positive improvements in employee motivation along with productivity levels. The theoretical foundations of behavioral economics also demonstrate successful long-term change resulting through nudging structured interventions embedded within a program. In fact, both researchers and practitioners of benefit from PBL methodologies and IKS-based wellness practices when combined in a structured organizational intervention programs.

This research has great importance because it connects the ancient knowledge with contemporary industrial requirements of construction business. Through strategic implementation of IKS and PBL organizations have the power to develop resilient staff members who demonstrate high satisfaction levels and contribute to long-term organizational growth, the research investigates effective strategies which policymakers can use with industry leaders and educators to establish employee care as an equal priority with workplace productivity and innovation.

3. Research Methodology

This research has been undertaken to integrate IKS into PBL for enhancing Life Satisfaction in Indian Construction Companies to gain actionable insights for policymaking. Researchers have opted for a pilot study method to evaluate the applicability of IKS principles in workplace well-being using a structured intervention.

Pilot studies serve as preliminary investigations that allow researchers to improve their research approaches and spot potential problems while collecting early evidence for subsequent research stages. As IKS-based professional intervention practices requires novelty to ensure that the systematic evaluation remains practical and impactful, the findings of the pilot study is expected to deliver important learning which will help develop extensive future interventions that address the requirements of Indian construction specialists.

The study involved 19 respondents from various roles within the Indian construction industry, including project managers, engineers, site supervisors, and administrative personnel. At 95% confidence with probability of a prevalent detectable problem as 15%, the adequate sample size for pilot study is 18.4 (<https://www.crutzen.net/n.htm>). The respondents were the participants of the intervention training program selected through purposive sampling, ensuring a diverse representation of job functions and experience levels.

The demographic profile included:

- Age range: 18–54 years
- Gender: Predominantly male, reflecting the industry's workforce composition
- Work experience: Predominantly more than 10 years in construction-related roles
- Education levels: Graduates, Postgraduates, and Doctorates in engineering, project management, and related fields of construction business

The structured intervention was an 8-hour workshop designed around key IKS principles, incorporating mindfulness, ethical decision-making, and holistic well-being practices. The session included:

- Introduction to IKS and Well-Being – Overview of Indian Knowledge Systems and their relevance to workplace resilience and mental health.
- Ayurvedic Perspectives on Stress Management – Practical applications of Ayurveda in diet, sleep, and work-life balance.
- Yoga and Pranayama for Workplace Productivity – Guided breathing exercises and simple asanas to enhance focus and reduce stress.
- Ethical Leadership and Decision-Making – Lessons from ancient Indian texts such as the Bhagavad Gita on professional ethics and self-regulation.
- Reflective Exercises and Group Discussions – Participants shared their workplace challenges and explored IKS-based solutions through case studies.

The intervention was designed to be interactive and experiential, ensuring participants could immediately apply the learnings in their professional environments. Data was collected through online self-reported surveys, allowing participants to share their experiences, perceptions, and behavioral changes anonymously. The surveys were structured as follows:

- Immediate Post-Intervention Survey: Participants' reflections on the following aspects of the questionnaire:
 - Physical and Mental Well-being
 - Relationship and Communication Skills
 - Purpose and Motivation at Work
 - Skills for Personal and Professional Growth
 - Financial and Time Management
 - Spiritual Growth and Control of Temperament
 - Overall Life Satisfaction
- Follow-Up Survey (90 Days Post-Intervention): Assessment of long-term impact, behavioral adaptation, and sustained well-being improvements for participants on aspects like:
 - Adherence to the Diet Plan
 - Adherence to the Workout Plan
 - Practicing Creative Pause
 - Experience of Sharing

- Actions taken to reduce personal weight
- Self-Reflection on Behavioural Changes

This time frame was selected to evaluate both short-term reflections and sustained behavioral transformations, providing a more comprehensive understanding of the intervention's effectiveness.

4. Findings and Discussion

Cronbach's Alpha is a measure of internal consistency, indicating how well the items within each dimension are correlated. A higher value (closer to 1) suggests greater reliability of the scale. Typically, thresholds used for interpretation are: ≥ 0.9 for excellent reliability, $0.8 - 0.89$ for good reliability, $0.7 - 0.79$ for acceptable reliability, and < 0.7 for questionable or poor reliability. Most dimensions exhibit good to excellent reliability, suggesting well-designed survey items (Table 1).

Table 1: Reliability Statistics of Immediate Post-Intervention Survey

Dimensions	No. of Items	Cronbach's Alpha
Physical and Mental Well-being	5	0.702
Relationship and Communication Skills	5	0.824
Purpose and Motivation at Work	5	0.903
Skills for Personal and Professional Growth	5	0.926
Financial and Time Management	5	0.869
Spiritual Growth and Control of Temperament	5	0.901
Leadership Skills	5	0.923
Networking Skills	5	0.921
Overall Life Satisfaction	10	0.762

Physical and Mental Well-being ($\alpha = 0.702$) represents acceptable reliability indicating that there might be scope to refine the items for better consistency. Relationship and Communication Skills ($\alpha = 0.824$) represents good reliability suggesting effectively measurement of this construct. Purpose and Motivation at Work ($\alpha = 0.903$) also represents excellent reliability indicating strong internal consistency. Skills for Personal and Professional Growth ($\alpha = 0.926$) also represents excellent reliability as the items within this dimension are highly consistent. Likewise, Financial and Time Management ($\alpha = 0.869$) has shown good reliability indicating well-correlated items. Further, Spiritual Growth and Control of Temperament ($\alpha = 0.901$) has shown excellent reliability that means high consistency in responses along with Leadership Skills ($\alpha = 0.923$) and Networking Skills ($\alpha = 0.921$). The final dimension of Overall Life Satisfaction ($\alpha = 0.762$) has shown acceptable reliability as this dimension has 10 items.

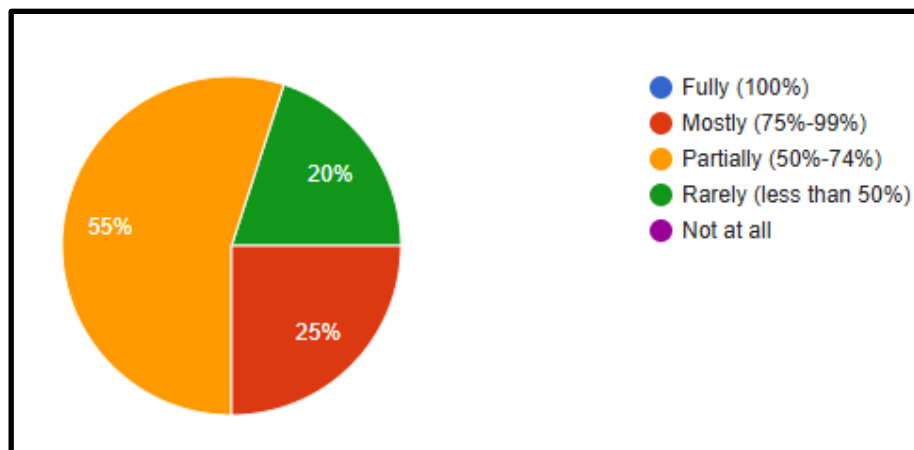


Fig. 1: Adherence to the Diet Plan

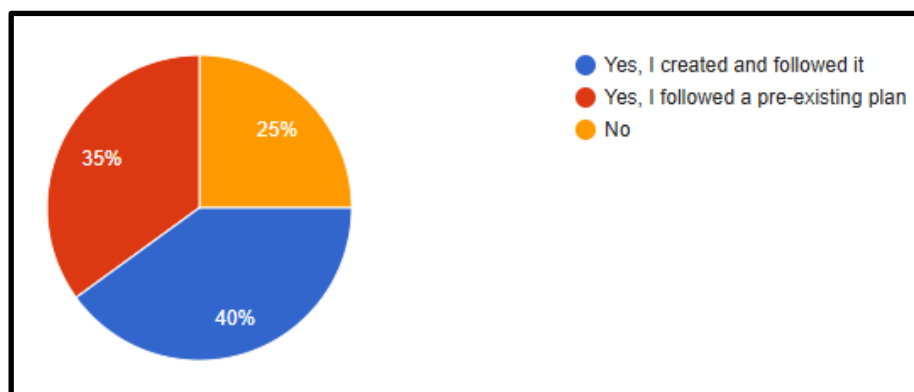


Fig. 2: Adherence to the Workout Plan

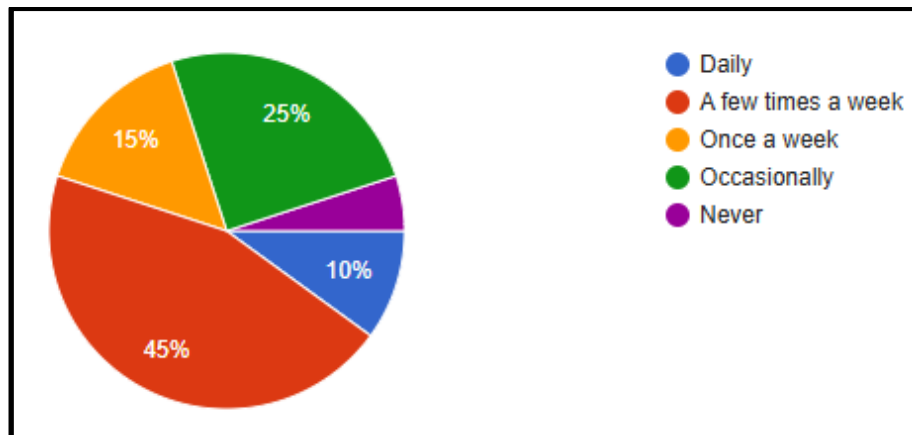


Fig. 3: Practicing Creative Pause

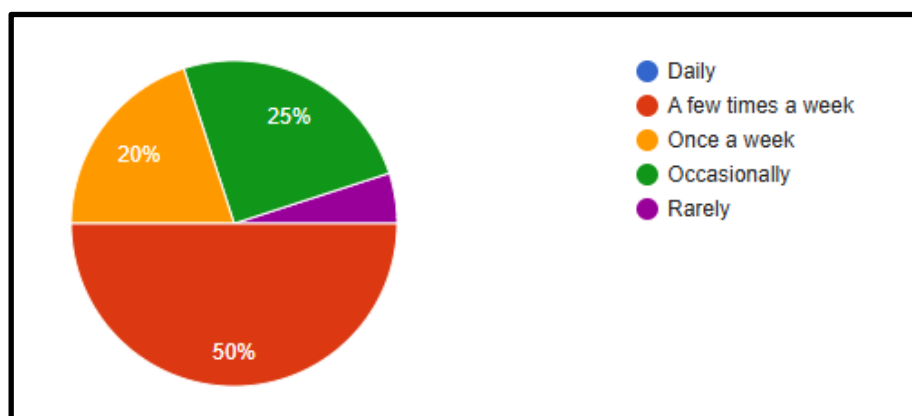


Fig. 4: Experience of Sharing

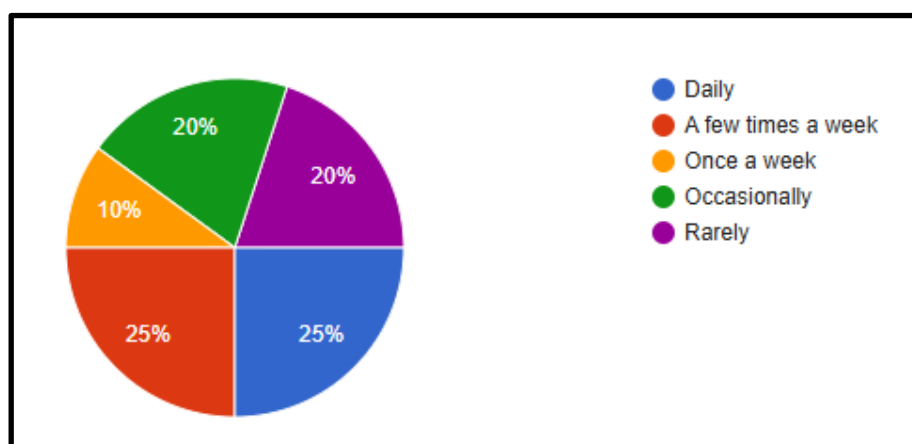


Fig. 5: Actions taken to reduce personal weight

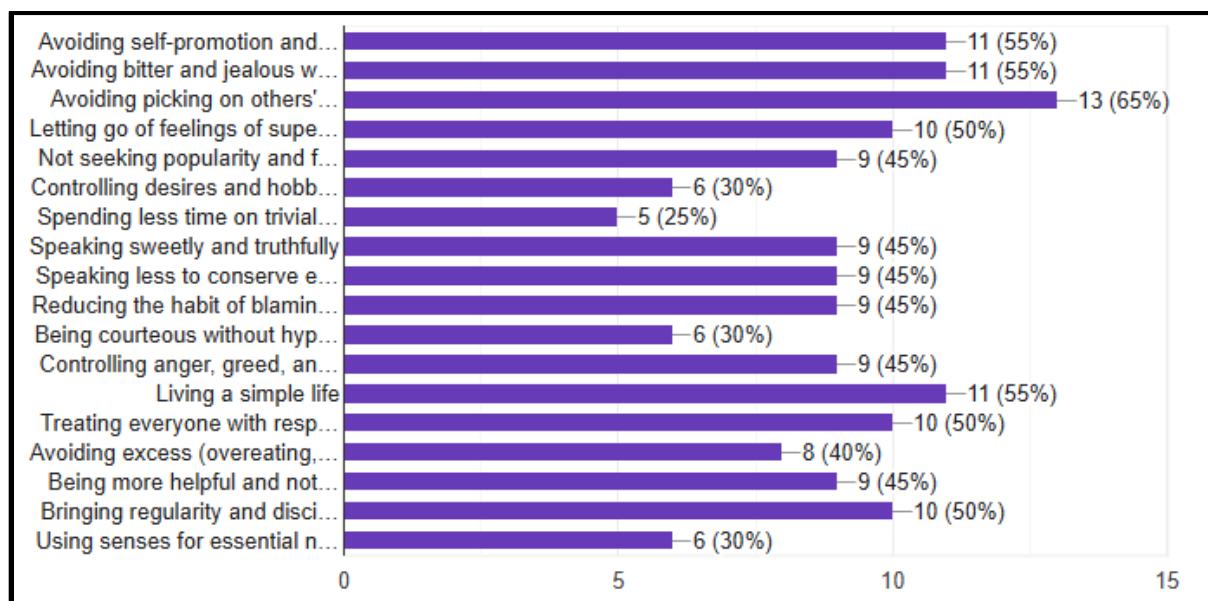


Fig. 6: Self-Reflection on Behavioural Changes

The reflections of Fig. 1, Fig. 2, Fig. 3, Fig. 4, Fig. 5, and Fig. 6 have been obtained in the Follow-Up Survey (90 Days Post-Intervention). The data presented in these figures clearly showcase the high internal consistency alongside the development of well-structured questionnaire suitable for full-scale deployment with minimal modifications for project-based learning. This successful pilot study serves as a preliminary assessment before full-scale research or program implementation.

5. Conclusion and Future Research

The Physical and Mental Well-being ($\alpha = 0.702$) and Overall Life Satisfaction ($\alpha = 0.762$) dimensions have lower internal consistency. This might require running an exploratory factor analysis (EFA) for these two constructs to verify whether all items truly belong to their assigned dimensions. Most dimensions have showcased strong internal consistency ($\alpha > 0.8$), which means that they are well-structured and they can be kept intact ensuring that they can be used as benchmarks for the study involving life satisfaction.

During the course of the intervention training program, participants agree to collaborate with trainers and take on the responsibility for their own learning through their own initiative and willingness. The trainers have played a successful role of facilitator while guiding, coaching, and offering opportunities and challenges for participants to push their own intellectual, psychological, social, and emotional boundaries.

The insights of this study will be useful in structuring the refinement for future surveys when planning full-scale implementation. Future studies will look for rewording complex statements or ensure that all items are phrased consistently. The future study will now involve expanded sample size with cross-sectional analysis, and also reflect on the long-term impact assessment of the intervention training program that involves experiential learning cycle for applications in project-based learning situations.

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